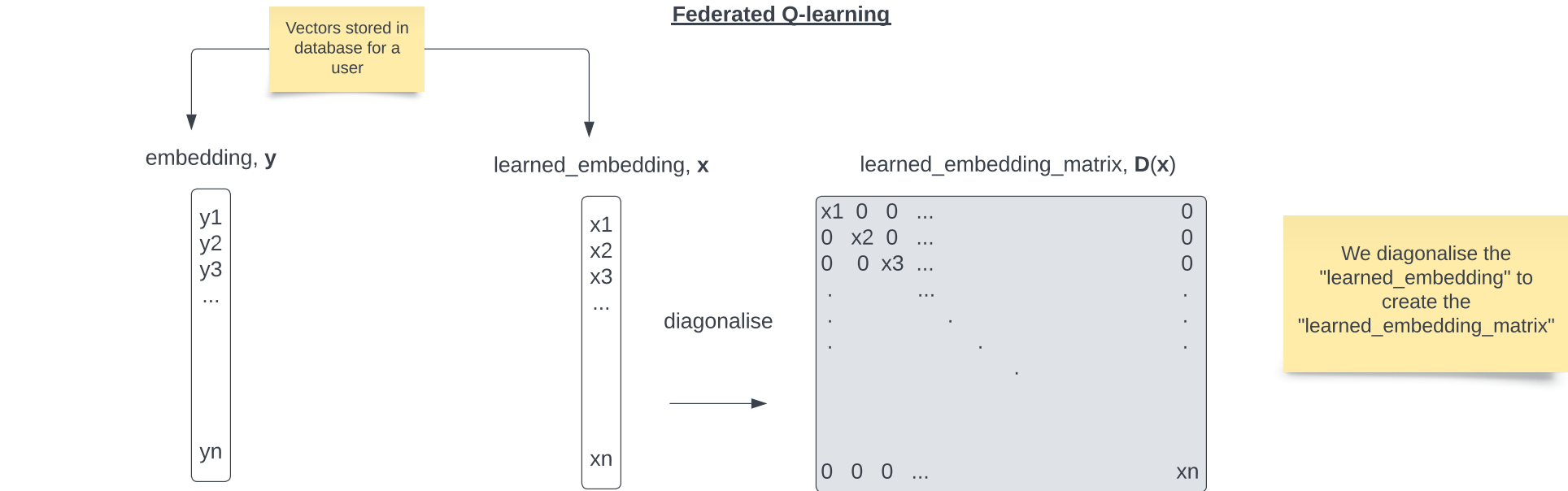
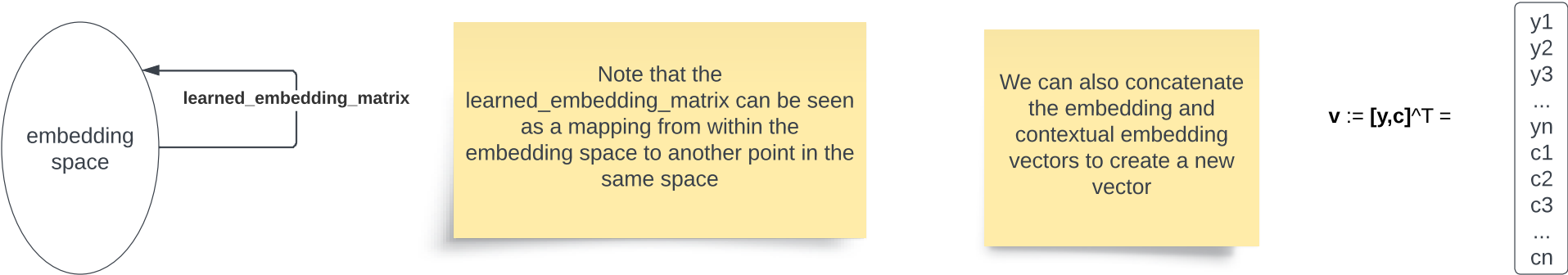
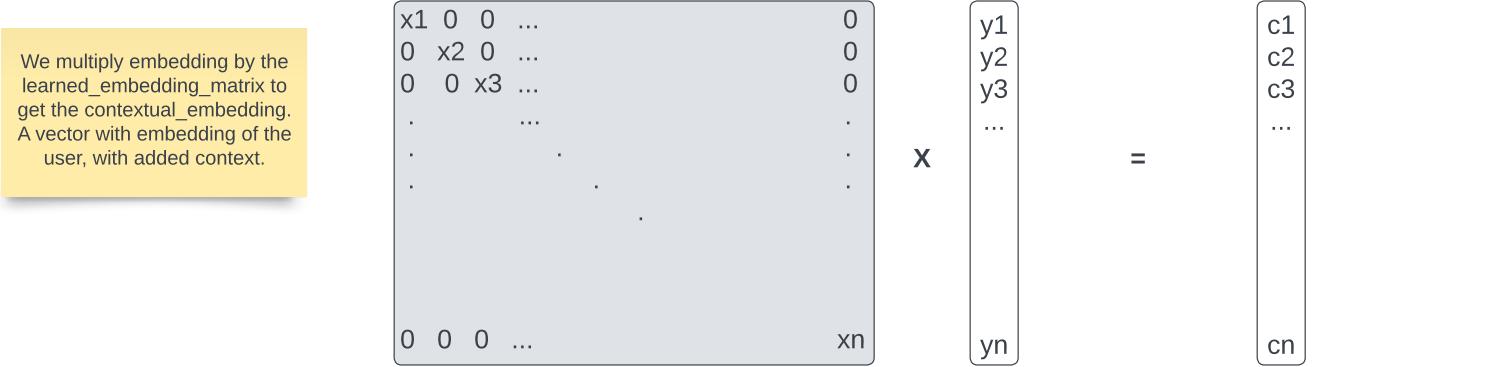


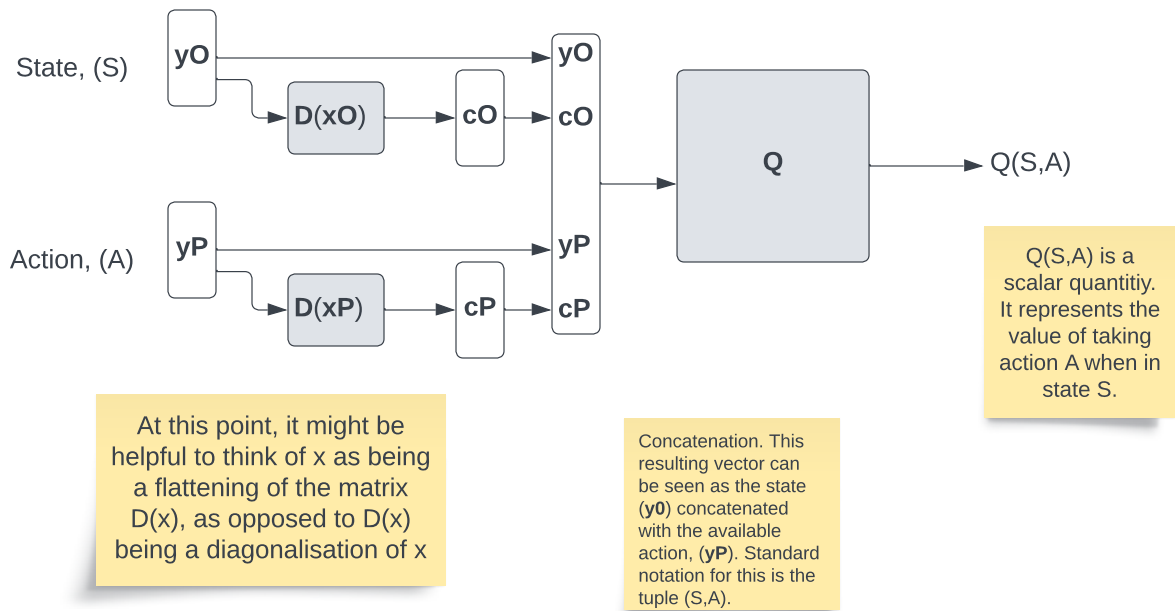
Federated Q-learning



let contextual_embedding, c , be a vector of the same rank as y
set $\text{contextual_embedding} := \text{learned_embedding_matrix} * \text{embedding}$



Let cards O and P be cards from the opportunities and profiles collections respectively. Let them have embedding vectors yO and yP , learned_embedding vectors xO and xP respectively. It may be helpful to note that O is reaching for, and we are evaluating the value of the action of presenting card P to O.



$L = Q(S,A) - R(S,A)$

We use back propagation to update the weights of Q and $D(x)$. $R(S,A)$ is the reward for taking action A in state S.

Note, the reason this sort of Q-learning can be seen as "Federated" is our use of vertical federation. We learn custom weights for each individual card, x . These weights are used to output a vector of new features, c . This vector c is a unique set of features (i.e.: a *feature expansion*) only available for this user's interactions. It is federated from the other training instances of the model. Federated learning is usually used to maintain data privacy. But, in our case we are using it to learn a contextual vector representaiton of a user's card, that is informed by the result of their actions and how their card is interacted with by others.

